Analysis of Customer Spending Behaviour: Ritlhavile Leol Makondo

An end-to-end data analytics projects that cleans, analyses, and visualizes a messy customer transection dataset to identify meaningful customer spending patterns and behaviours that can support informed business decision making.

Pipeline: Excel (Data cleaning) →SQL/MYSQL (Data analysis) → Power BI (Data visualisation

Background

Understanding customer behaviour is a central to how business make decisions about marketing, retention, and product strategy. In practice, the data available for this kind of analysis is rarely clean, it usually contains duplicate records, missing values, inconsistent formatting, and outliers that need to be identified and resolved before any reliable analysis can take place.

This project focuses on customer transactions dataset that contains numerical data representing customer spending patterns, incomes, ages, and purchase behaviours. The aim is to identify meaningful pattens that could support informed business decision making.

Aim and objectives

Aim: To identify meaningful spending patterns and behaviour that can support informed business decision making.

Objectives:

- To clean and prepare the customer transection dataset.
- To explore spending pattens based on age, income and purchase frequency.
- To identify customer group with similar spending behaviours.
- To generate meaningful insight that can help support business decision-making

Data

Source: Kiggle – Size: ~10,000 rows, 6 columns

Data quality issues identified:

Issue	Details
Duplicate rows	200 rows were identical across every column
Missing values	247 missing Age values, 242 missing Transection Amount
Outliers	6 unrealistic Age values (0-3 years old)
Invalid values	218 negative Transection_Amount values, with no refund/return field to explain them
Unreliable ID values	Repeated Customer_ID values often had different Age/Income attached

After data cleaning, the dataset was reduced to 9,989 rows with no missing values or duplicates records.

Methodology

Phase 1: Data cleaning (Excel)

- Removed exact duplicate rows
- Documented Customer_ID as unreliable and treated each row as an individual transection record rather than unique customer
- Imputed missing Age and Transection_Amout values using the column median (chosen over the mean to avoid distortion from outliers)
- Investigated and removed unrealistic Age values (Age 0-3) and negative Transection _Amount entries
- Addes derived columns, Age_Group and Income_Bracket to support later segmentation

Phase 2: Analysis (SQL/MYSQL Workbench)

- Imported the cleaned dataset into a MySQL database
- Converted Annual_Income and Transection_Amount from formatted text into numeric field using SQL string function
- Wrote queries to group and segment customer by income, spending score, and purchase behaviour

Phase 3: Visualization (Power BI)

- Connected Power BI to the clean dataset
- Built an interactive dashboard to explore customer segments and spending pattern

Results

Table 1

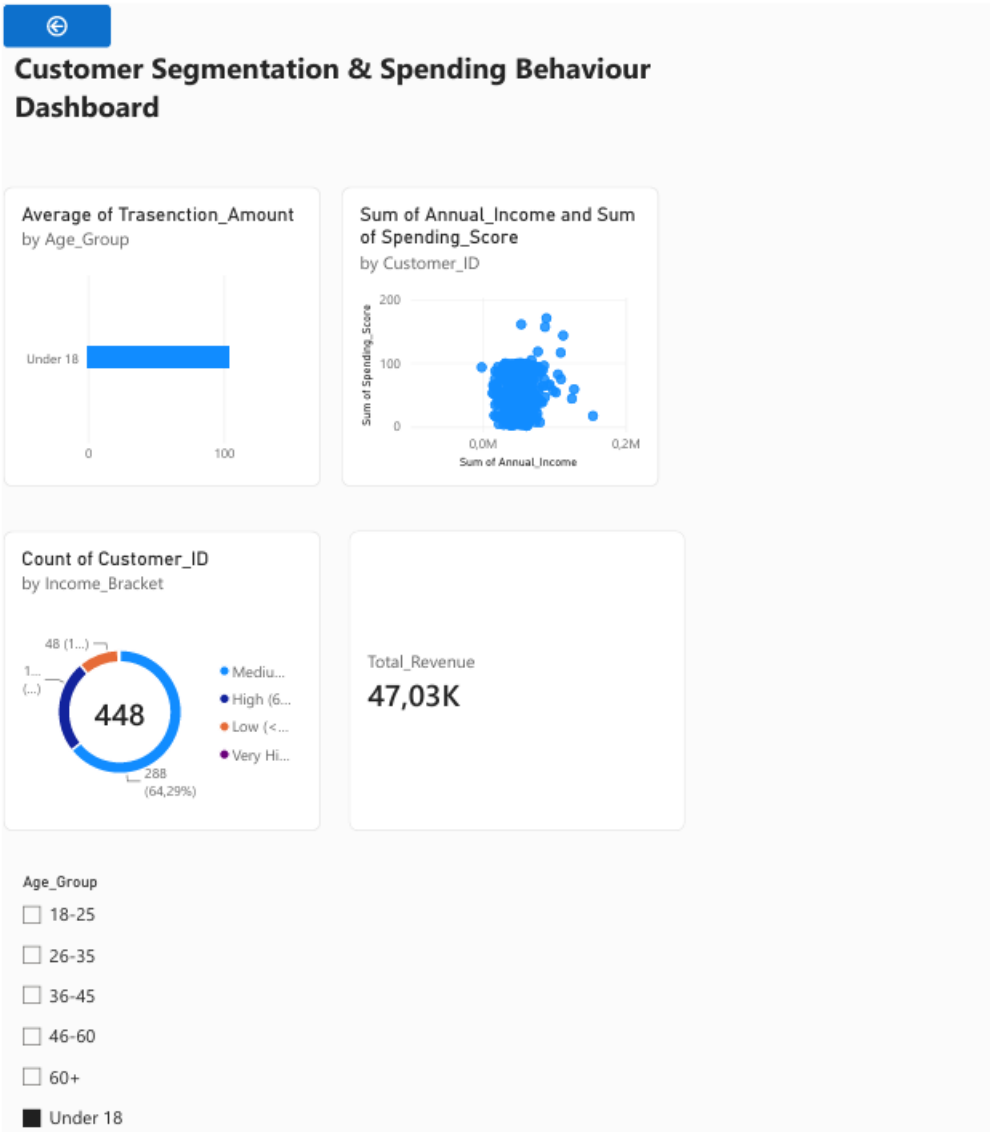
Age group	Average Spending	Number of customers
<18	104.98011607	448
18-25	99.05964000	1380
26-35	101.46632352	3686
36-45	101.46748160	3179
46-60	100.36291785	1255
>60	95.80019512	41

Table 1 present average spending scores alongside customers counts across six age groups, ranging from under 18 to over 60 years old. The results indicate relatively consistent spending behaviour across most segments, with average scores clustering between approximately 99 and 101.5 for the 18-25 through 46-60 age groups. The under 18 segments recorded the highest average spending score at 104,98, while the 60+ segment recorded the lowest 95.80, representing the only two groups that deviate meaningfully from the overall patten. Customer distribution is heavily concentrated in the 26-35 and 36-45 age groups, which together account for 6,865 of the 9,989 customers, in contract to 41 customers in the 60+ category. Taken together, the data suggest that average spending does not vary substantially by age, indicating that age alone is unluckily to be the strong determinant of spending behaviour withing this dataset.

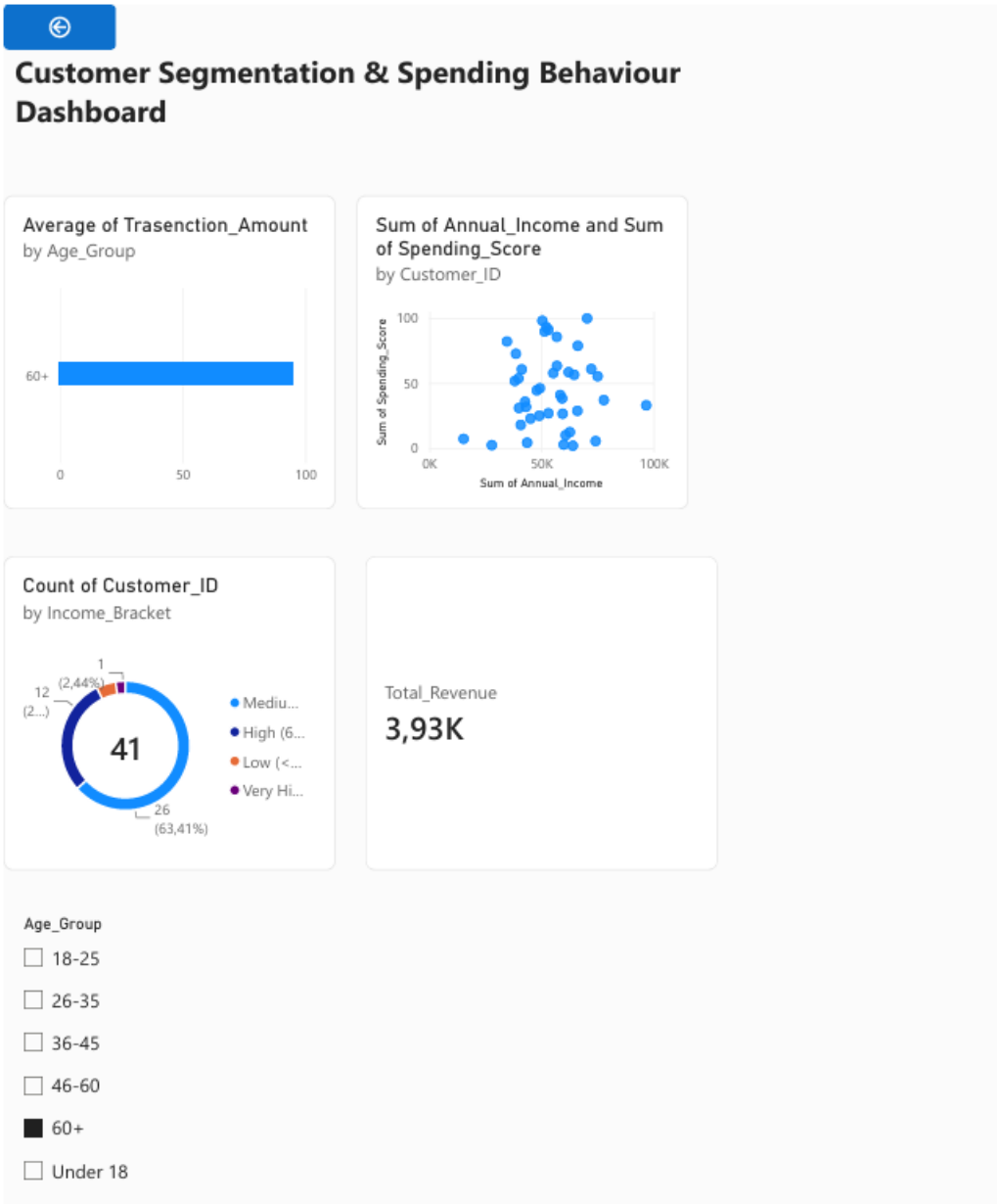
Table 2

Income bracket	Average score
Low (<30k)	50.312145
Medium (30k-60k)	50.684291
High (60k-90k)	50.546535
Very high (90k+)	47.512821

Table 2 presents average spending scores across four income brackets, ranging from low (under 30k) to very high (90k and above). Scores remail consistent across low, medium and high-income brackets, clustering narrowly between 50.31 and 50.68. the vary high income brackets is only clear outlier, recording a noticeable lower average score of 47.51 compared to the other three groups. Overall, this suggests that higher income does not correspond to higher spending scores in this dataset, indicating that income alone is unlikely to be strong predictor of spending behaviour.



Average spending, income vs spending score, and income bracket split filtered to under 18(448 customers, R47,03k revenue), showing noticeable denser income vs spending cluster.



Average spending, income vs. spending score, and income bracket split filtered to the age 60+ age group (41 customers, R3,93k total revenue)

Conclusion

This analysis shows that spending behaviour does not vary meaningfully by age or income, meaning neither factor reliable predicts customer value in this dataset. As a result, it is recommended that marketing and retention strategies avoid segmenting customers by age or income bracket alone, since doing so would not effectively distinguish higher-value customers from lower-value ones. Instead, other variables such as purchase frequency should be explored as potentially stronger indicators of spending behaviour. Given the small sample size in the 60+ segment, it is recommended that future data collection aim for a more balanced representation across age groups before drawing conclusions from the category. Overall, this project demonstrates the ability to clean, process, and analyse data through a complete pipeline, while also recognizing when the data does not support strong demographic-based conclusions.

Teck Stack

- **Excel –** Data exploration and cleaning
- **MYSQL –** Data transformation and segmentation queries
- **POWER BI –** Interactive dashboard and visualisation

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